Note

Recall that subtraction on numbers is not commutative. Is set difference commutative? We expect that the set difference is not commutative as well. For instance, consider $A = \{a, b, c\}$, $B = \{b, c, d\}$. $A - B = \{a\}$, $B - A = \{d\}$; we see that $A - B \neq B - A$.

1.6.2 Associative Property

Now, we perform operations on union and intersection for three sets.

Let $A = \{-1, 0, 1, 2\}$, $B = \{-3, 0, 2, 3\}$ and $C = \{0, 1, 3, 4\}$ be three sets.

Now, $B \cup C = \{-3, 0, 1, 2, 3, 4\}$

$A \cup (B \cup C) = \{-1, 0, 1, 2\} \cup \{-3, 0, 1, 2, 3, 4\}$

$= \{-3, -1, 0, 1, 2, 3, 4\}$ ... (1)

Then, $A \cup B = \{-3, -1, 0, 1, 2, 3\}$

$(A \cup B) \cup C = \{-3, -1, 0, 1, 2, 3\} \cup \{0, 1, 3, 4\}$

$= \{-3, -1, 0, 1, 2, 3, 4\}$ ... (2)

From (1) and (2), $A \cup (B \cup C) = (A \cup B) \cup C$.

This is associative property of union among sets A, B, and C.

Now, $B \cap C = \{0, 3\}$

$A \cap (B \cap C) = \{-1, 0, 1, 2\} \cap \{0, 3\}$

$= \{0\}$ ... (3)

Then, $A \cap B = \{0, 2\}$

$(A \cap B) \cap C = \{0, 2\} \cap \{0, 1, 3, 4\}$

$= \{0\}$ ... (4)

From (3) and (4), $A \cap (B \cap C) = (A \cap B) \cap C$.

This is associative property of intersection among sets A, B and C.

Associative property: For any three sets A, B and C

(i) $A \cup (B \cup C) = (A \cup B) \cup C$ (ii) $A \cap (B \cap C) = (A \cap B) \cap C$

Example 1.20 If $A = \left\{-\frac{1}{2}, 0, \frac{1}{4}, \frac{3}{4}, 2\right\}$, $B = \left\{0, \frac{1}{4}, \frac{3}{4}, 2, \frac{5}{2}\right\}$ and $C = \left\{-\frac{1}{2}, \frac{1}{4}, 1, 2, \frac{5}{2}\right\}$, then verify $A \cap (B \cap C) = (A \cap B) \cap C$.

1-SET LANGUAGE.indd 21 02/12/2022 21:30:59